Efficient Convolutional Neural Architecture Search for Remote Sensing Image Scene Classification

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***Abstract*— As a fundamental but challenging task in the interpretation of remote sensing images, scene classification plays an important role in various applications and has become an active research topic. Many previous works have demonstrated the remarkable performance of the deep convolutional neural networks (CNNs) for remote sensing scene classification. However, the progress made by CNN-based methods for scene classification has gradually reached saturation in recent years, due to the serious dependence on the pretrained CNN models, the limitations of manually designed network architecture and the disadvantages of existing data sets. In this article, a new paradigm to automatically design a suitable CNN architecture for scene classification is investigated. We propose an efficient architecture search framework to discover optimal network architectures in continuous search space with the gradient-based optimization method. Our framework consists of two stages: the search phase and the evaluation phase. During the search process, a greedy and progressive search strategy is introduced to search network building blocks (i.e., cells) through bilevel optimization. Besides, we propose a simple architecture regularization scheme to further improve the search efficiency and the robustness of discovered architectures. After the search process, the optimal cell architectures are determined and then repeatedly stacked to construct the final network for evaluation. For the data set, we propose a mergence strategy to build a new large-scale remote sensing scene image data set that contains rich scene categories and image diversity, making it feasible to find a new CNN model with strong generalization ability for scene classification. Extensive experiments demonstrate the efficiency of the proposed search strategies and the impressive classification performance of searched CNN architectures on seven public benchmark data sets, including four large-scale data sets and three small-scale data sets.**

***Index Terms*— Convolutional neural networks (CNNs), neural architecture search (NAS), remote sensing image, scene classification.**

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I. Introduction

REMOTE sensing image analysis, which refers to understand and interpret the semantic content of the ground surface, has become an active research topic in the community of Earth observation and raised increasing interests due to the rapid development of Earth and space observation technologies [1], [2]. In recent years, a massive number of remote sensing images with very high spatial resolution and complex spatial patterns have become more accessible, which offers unprecedented conveniences and opportunities for many research directions, such as remote sensing scene classification [3], change detection [4], [5], geographic image and video retrieval [6], land-use classification [7], [8], and automatic target recognition [9]. As a fundamental but important problem among the field of remote sensing images interpretation, scene classification aims to assign each unlabeled aerial image a specific semantic label (e.g., airplane, river, highway, and farm) based on the semantic content of it. This task has widespread applications, including environmental monitoring, residential planning, land resource management, etc. [10], [11]. However, the remote sensing scene images usually contain various types of land-cover or ground objects which vary in shapes, scales, orientations and spatial distributions. Scene images with the same semantic labels may exhibit high discrepancies in the geometrical structures and spatial patterns, or share very few spatial similarities [12]. For example, urban areas may consist of many different geographic structures such as residential areas, roads, and commercial areas, while they are different semantic classes. Therefore, the low interclass discrepancy and high within-class variability make scene classification a considerably challenging task.

Technically speaking, the task of remote sensing scene classification generally consists of two basic parts: feature extraction and classification. In terms of developing robust and discriminative feature representations for scene images, most traditional approaches exploit supervised classifiers that are trained on a set of specific handcrafted features (e.g., scale invariant feature transform (SIFT) [13]) which are encoded by various local feature descriptor encoding methods [e.g., bag-of-visual-words (BoVWs) model [14], spatial pyramid matching (SPM) method [15], and Fisher Vector (FV) [16])]. These methods, generally called low-level methods, attempt to describe the image appearance and content through low-level semantic features (e.g., color, textures, or edges). However, the performance and potential of these approaches are

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seriously limited by goal-specific feature selection or design approaches, which are commonly selected by human experts.

In recent years, the deep convolutional neural networks (CNNs), a powerful end-to-end global feature extraction framework, has achieved a remarkable breakthrough and obtained state-of-the-art performance in many images understanding tasks [17], [18]. Due to the powerful ability to automatically learn complicated and relevant contextual features, various CNN models have also been widely adopted in the task of remote sensing scene classification and have gradually become mainstream methods [19]. The early CNN-based methods for scene classification prefer to fully train a new CNN model from scratch, however, the CNN models are a set of data-driven methods, and a considerable amount of labeled data is necessary for training a deep CNN that usually contains millions of parameters [20]. Unfortunately, compared to the natural images, most available remote sensing image data sets contain very limited well-annotated data, making it infeasible to fully train a deep CNN model for a specific task. To address this issue, various pretrained CNNs based on ImageNet (e.g., CaffeNet [21], VGG [22], and GoogLeNet [23]) have been widely adopted in recent works. These methods are inspired by transfer learning and the transferred high-level features have proven to be effective for scene classification tasks without any adaption modalities (finetuning or training from scratch). In addition, some recent works demonstrate that the pretrained CNN model can effectively alleviate the overfitting problem and is suitable to be used as feature extractors to construct discriminative representation for scene classification in the absence of sufficient training data [24]. Actually, transferring the feature representation from a pretrained deep CNN to the target task has become a standard practice in the field of remote sensing scene classification.

However, a recent study [25] revealed that ImageNet pretraining-based methods may not be the optimal solution to the task that usually has less training data. The authors demonstrated that ImageNet pretraining does not necessarily help avoid overfitting and does not automatically provide better regularization than random initialization. On the other hand, the time and computation cost of ImageNet pretraining are often ignored when researching the target task, which leads to the wrong impression that using pretraining can reduce the time cost. Furthermore, there are significant and special differences between aerial images and natural images, for example, the remote sensing scene images are usually local patches extracted from large satellite images with different resolution, which consist of highly complex geometrical structures and the ground objects vary greatly in view angles, scales, and shapes. Almost all of the available CNN-based methods for scene classification are originally designed based on natural images and have not been improved to adapt to the specific spatial patterns of remote sensing images, which limits the potential of deep learning methods and affects the classification performance. Therefore, it is clear that there is an urgent demand to design a dedicated CNN model for scene classification instead of directly construct feature representation from pretraining CNN models.

The design and construction of the neural network architecture have proven to be significant, and many advances in deep learning stem from its immediate modifications. As time-consuming engineering of trial and error, most of the existing CNN architectures for image classification are designed manually through experiments and expert experience, which also requires a high level of domain knowledge. Most recently, there has been increasing interest in neural network architecture search methods that aims to automate the architecture design process. Neural architecture search (NAS) [26], an emerging and fast-developing research direction in the automated machine learning community, has achieved impressive empirical performance in various tasks including image classification. However, early NAS approaches are still time-consuming and computationally expensive, requiring to explore and evaluate thousands of architectures on the target data set in a single experiment. In addition, most of the existing NAS are performed on natural images analysis, few works of literature have reported the performance of NAS in the task of remote sensing images interpretation that relies on very high-resolution images, multidomain, and multiscale scene representation.

For the task of remote sensing scene classification, tremendous efforts have been made to obtain more discriminative feature representation from the pretrained CNN models. However, the classification performance based on pretrained CNN models has gradually reached saturation due to the limitations of the high-level features extracted from the pretrained models and the notable disadvantages of existing remote sensing image data sets. In this work, we explicitly tackle the performance bottlenecks in the task of remote sensing images scene classification. We investigate a new paradigm to automatically design a suitable CNN architecture for scene classification. The main contributions of this article are presented as follows.

1) We investigate a new paradigm to automatically design a suitable CNN architecture for scene classification. A greedy and progressive architecture search (GPAS) framework is presented, which aims to discover optimal CNN architecture in continuous search space with the gradient-based optimization method. The core idea is to progressively reduce the search space and increase the network depth during the search process by introducing a greedy strategy for path-level pruning, which guarantees a much high correlation between the search and evaluation phase.
2) We propose an efficient architecture regularization scheme to alleviate over-fitting and further improve the efficiency of search process. Specifically, the paradigm of "early stopping" and partial training are introduced. In the early search, the number of search epochs is small and only part of the training data is used to optimize the architecture parameter, which aims to obtain a rough ranking of the edge importance. We progressively increase the number of search epoch and training data to learn a more stable ranking of architecture parameters.
3) We propose a mergence strategy to build a new large-scale remote sensing scene image data set, referred to as

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very-high-resolution remote-sensing (VHRRS) data set, which combines multiple public scene data sets with different domains. The merged data set contains rich scene categories and image diversity, making it feasible to find a new CNN model with strong generalization ability for scene classification.

4) We demonstrate the effectiveness of the proposed architecture search method on several public benchmark data sets, including four large-scale data sets and three small-scale data sets. Extensive experiments show that the discovered architecture achieves better classification results compared with the full-training CNN model and pretrained CNN-based methods. Moreover, the proposed framework performs better than the other four representative NAS methods in search efficiency and stability.

The remainder of this article is organized as follows. Section II provides a brief introduction to the related works, including various deep learning-based methods for scene classification and the development of NAS methods. In Section III, we first give a brief description of the principles of differentiable architecture search methods, then we provide a detailed explanation of our proposed methods. Sections IV and V deal with the experiments setup and the classification results on seven scene data sets. Our conclusions are presented in Section VI.

II. Related Work

Remote sensing scene classification is an active research topic in the community of remote sensing image interpretation. Over the past years, numerous algorithms have been developed for better performance, in which CNN-based methods have become the most popular methods due to their powerful ability to automatically extracting features from unstructured data. On the other hand, the choice of CNN architecture has significant effects on the performance. NAS, aims to automatically design neural network architectures, is a new research direction in automated deep learning. Recent NAS methods have achieved impressive results and have been applied in many tasks, including image classification. In this section, we first provide a brief introduction to various deep learning-based methods for scene classification, then we summarize the development of NAS methods.

A. Deep Learning Methods in Scene Classification

The breakthrough made by deep learning in image understanding tasks is largely due to its automation of feature extraction: hierarchical feature representations are learned in an end-to-end fashion from raw data rather than designed by human experts. Most recently, many works have shown that deep learning methods have achieved impressive progress in remote sensing scene classification. We summarize the existing deep learning-based methods for scene classification into five categories:

1) Directly exploiting pretrained CNN models, which refers to transferring the pretrained CNNs (e.g., AlexNet [27] GoogLeNet [23]) on the natural images to scene classification tasks and performing fine-tuning of the parameters. Because the available training data are always insufficient, these methods have proven to be effective in early works. For instance, He *et al.* [28] utilized the fully connected layer of the pretrained VGGNet-16 and GoogLeNet models to extract the remote sensing scene representations. Cheng *et al.* [29] attempted to fine-tune the existing AlexNet, GoogLeNet, and VGGNet-16 to obtain competitive performance.
2) Using the deep CNN as a local feature extractor and combine it with unsupervised feature coding approaches. Different from directly exploiting the features from the pretrained CNN models as the final representation, these methods reprocess the extracted feature to adapt the target data set. Zhao and Du [30] constructed a well-designed CNN model to extract local spatial features from multiscale patches and then the local spatial patterns are encoded into a scene representation with the BoVWs. Zheng *et al.* [10] performed a multiscale pooling on the last convolutional layers to extract the local features and utilized the FV method to generate the holistic scene representation.
3) Training new networks from scratch. The weights of a new or existing network are all randomly initialized and then trained on the target data sets. This approach is preferable since it considers the discrepancies between remote sensing and natural images, which tends to yield a more robust network. Zhang *et al.* [31] proposed a gradient boosting random convolutional network (GBRCN) combining different deep CNNs to classify remote sensing images. To obtain better and specific feature from small-scale scene data sets, Liu and Huang [32] proposed a triplet network for weakly supervised training.
4) Using various fusion technologies. Through different levels of fusion strategies including feature, model or raw data, these methods usually capable of constructing more discriminative feature representation. Zhu *et al.* [33] proposed two effective feature fusion approaches to fuse the local and global semantic features. Zhao *et al.* [34] incorporated the multispectral information and the structural feature hidden in scene images to increase the performance. Du *et al.* [35] presented a fusion strategy to construct the feature representation by combining multilayer activations of a single pretrained CNN model and extend it to an ensemble framework of multiple CNN models.
5) Introducing other new strategies to further improve the performance such as metric learning [36], attention mechanism [37], semisupervised generative models [38], loss function improvements [20], and so on. These methods aim to address several specific issues, such as multiscale scene representation, cross-domain classification, within-class diversity, and interclass similarity.

B. NAS Methods

NAS has attracted lots of attention in recent years. It has changed the design convention of model architecture from manual to automatic and has achieved promising performance in deep learning community. The goal of NAS methods can

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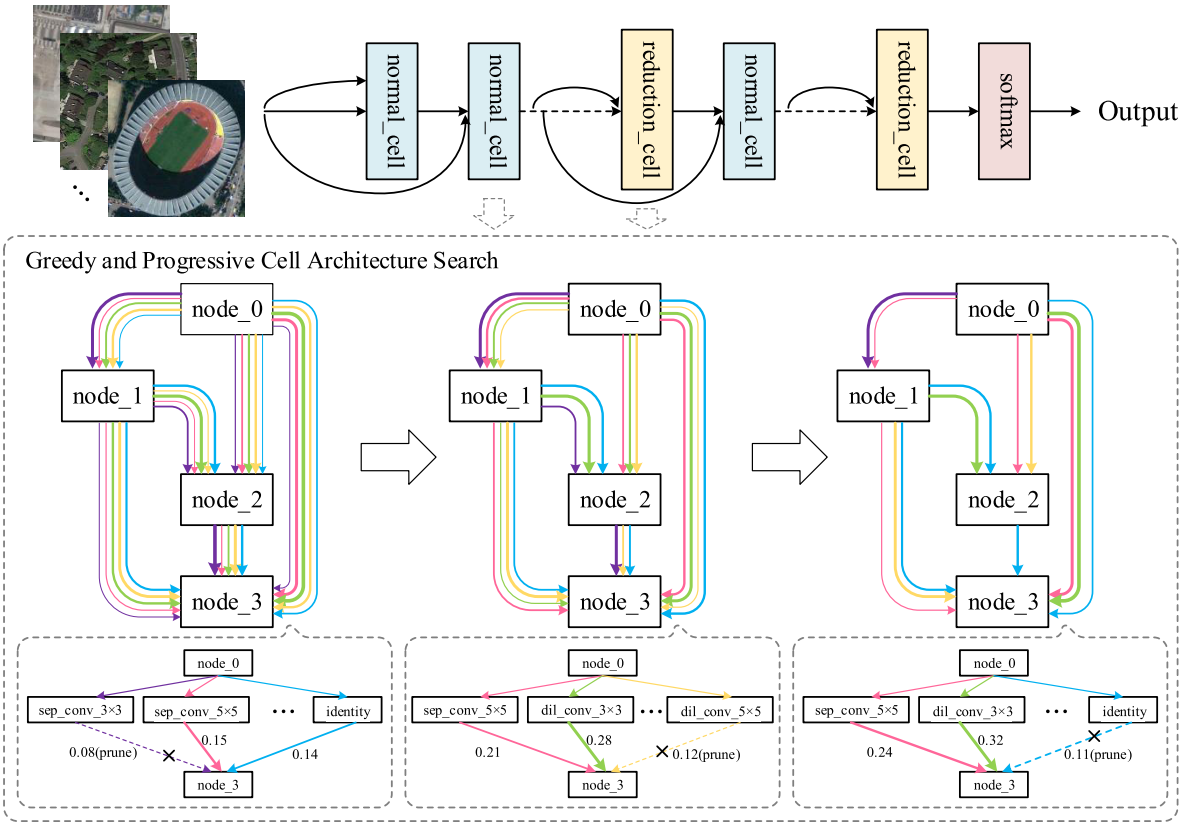


Fig. 1. Overall framework of our method for scene classification. A batch of raw scene images are used as the inputs of an automatically designed CNN model, which contains multiple normal cells and reduction cells. The architecture of each cell is discovered through a greedy and progressive search strategy.

be described as discovering the optimal architecture in a predefined search space so that the validation accuracy is maximized on the target task. The early works of NAS mainly focused on the optimization of network hyperparameters, such as the number of layers, filter size, and the type of activation function. The work [39] marks the beginning of research on the automatic search for network topologies. Since then, many NAS algorithms have been investigated. Black-box optimization based on reinforcement-learning (RL) approach is the first proven NAS method with state-of-the-art performance on image classification. Baker *et al.* [40] proposed a Q-learning agent to search neural network architectures. The agent adopts a ϵ-greedy policy: with probability $1-\epsilon$, it selects the action that results to the best expected return (i.e., accuracy) estimated by the current model, otherwise uniformly selects an action. Zoph and Le [41] built an RNN agent trained with Policy Gradient to search CNN and LSTM architectures. Shortly thereafter, the work [42] demonstrated that similar results could also be achieved by Evolutionary Algorithm (EA)-based method, where each neural network architecture is represented as a string of genes and search the architecture space by mutation and recombination. However, both RL-based and EA-based approaches are computationally expensive and time-consuming (e.g., hundreds or even thousands of GPU-days) as they search and evaluate numerous architectures from scratch in discrete search space. A novel search strategy for the architectures over a continuous domain was proposed by DARTS [43], which relaxes the search space to be continuous, making architecture search differentiable and much more efficient. This method has opened up a path toward computationally feasible architecture search. Until now, most of the existing works on NAS focus on the natural image classification task and there are a few works of literature reported the ability of NAS in the remote sensing field. In this work, we investigate a new paradigm to design suitable architecture for scene classification through the NAS method and demonstrate the potential of that in remote sensing image interpretation.

III. Methodology

In this section, we first introduce the principles of differentiable architecture search methods and analyze the limitations of DARTS. Then we describe the proposed search framework for remote sensing scene classification, where a greedy and progressive search strategy is introduced to strengthen the correlation between the search and evaluation phase. In addition, we propose an effective architecture regularization scheme to alleviate over-fitting and generate robust models. The overall framework of our method for scene classification is illustrated at Fig. 1.

A. Principles of Differentiable Architecture Search Methods

As a gradient-based approach for architecture search, DARTS has produced high-performance architectures in many tasks. Compared to RL-based and EA-based methods, DARTS

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is able to find good network architectures with a search cost of a few GPU-days.

Following the works [26], [39], DARTS searches for a cell-based space, where a normal cell and a reduction cell is repeatedly constructed to build up the final architecture. In a cell, a small directed acyclic graph (DAG) is used to represent its topology architecture, consisting of N ordered intermediate nodes. Each node is a latent representation and denoted as $x^{(i)}$, where i is its topological order in the DAG. Each directed edge $e^{(i,j)}$ in the DAG is associated with a candidate operation $o^{(i,j)}$ that transfers the information from node $x^{(i)}$ to node $x^{(j)}$. The intermediate node $x^{(j)}$ is computed based on all of its preceding nodes

$$x^{(j)} = \sum_{i<j} o^{(i,j)}(x^i). \tag{1}$$

Let $\mathcal{O} = \{o_1^{(i,j)}, o_2^{(i,j)}, \ldots, o_M^{(i,j)}\}$ be a set of M candidate operations (e.g., convolution, max pooling, and identity) on edge $e^{(i,j)}$. In DARTS, the optimal architecture is derived from a discrete search space by relaxing the selection of operations to a continuous optimization problem. During the search phase, the operation of each edge is parameterized by architecture parameters $\alpha^{(i,j)}$ as a softmax mixture over all the possible operations within the operation space $\mathcal{O}$

$$\bar{o}^{(i,j)}(x) = \sum_{o\in\mathcal{O}} \frac{\exp(\alpha_o^{(i,j)})}{\sum_{o'\in\mathcal{O}} \exp(\alpha_{o'}^{(i,j)})} o(x) \tag{2}$$

where each operation $o^{(i,j)}$ is associated with a continuous coefficient $\alpha_o^{(i,j)}$, and $\alpha^{(i,j)}$ is a vector with dimension $|\mathcal{O}|$ that indicates the mixing of operation between a pair of nodes. After relaxation, the architecture search phase jointly performs learning on a set of continuous variables $\alpha = \{\alpha^{(i,j)}\}$ and the network weights w within all the mixed operations. For the search procedure, we denote $\mathcal{L}_{\text{train}}$ and $\mathcal{L}_{\text{val}}$ as the training and validation loss, respectively. The architecture search problem is reduced to learning α^* that minimizes the validation loss $\mathcal{L}_{\text{val}}(w^*, \alpha^*)$, where the value of network weights w is obtained by minimizing the training loss $\mathcal{L}_{\text{train}}$. DARTS resolves this problem with a bilevel optimization, where α is the higher level variable, and w acts as the lower-level variable

$$\min_{\alpha} \ \mathcal{L}_{\text{val}}(w^*(\alpha), \alpha) \tag{3}$$

$$\text{s.t. } w^*(\alpha) = \arg\min_{w} \mathcal{L}_{\text{train}}(w, \alpha). \tag{4}$$

After architecture parameters α are obtained, the final discrete architecture is derived by: 1) setting $o^{(i,j)} = \arg\max_{o\in\mathcal{O}, o\neq\text{zero}}(\exp(\alpha_o^{(i,j)}))/(\sum_{o'\in\mathcal{O}} \exp(\alpha_{o'}^{(i,j)}))$, and 2) for each intermediate node, choosing two incoming edges with the two largest values of $\max_{o\in\mathcal{O}, o\neq\text{zero}}(\exp(\alpha_o^{(i,j)}))/(\sum_{o'\in\mathcal{O}} \exp(\alpha_{o'}^{(i,j)}))$.

Despite the efficiency of DARTS, there are still some limitations should be noticed. First, it aggregates multiple features generated by all candidate operations and updates all α and w in one step, which results in expensive computational cost because a large number of intermediate calculation values need to be saved in memory. Therefore, the GPU memory usage is proportional to the depth of the super-network, which forms a major obstacle to direct applications. On the other hand, it contains two stages: a search phase and an evaluation phase. During the search phase, it searches the architecture with shallow layers (8 cells), while the searched cell architecture is stacked to a deeper network (20 cells) in the evaluation phase, there is a big discrepancy between the performance of shallow and deep networks. The best ranking derived from the search phase does not imply the actual ranking in the final evaluation, which is referred as a domain-shift problem [44]. To address these issues, we proposed a novel and efficient architecture search framework, which is described in more detail in Section III-B.

B. GPAS Framework for Scene Classification

Recent advances in NAS show a lot of promise, especially the differentiable architecture search methods, which reduce the search cost from thousands of GPU days to a few GPU days. This framework has opened up a new path toward scene classification, that is to automatically design CNN models employing a computationally feasible architecture search method. In this section, the proposed GPAS methods for scene classification are described.

1) Continuous Relaxation of Cell-Based Search Space: The search space includes the network topology that can be represented in principle. Appropriate search space can reduce the computational cost and simplify the search process. Upon observation that various effective handcrafted architectures are based on repeating a fixed block, a cell-based search space is constructed in our work. Each cell is regarded as a DAG which consists of several edges and N ordered intermediate nodes. We denote each node as $x^{(i)}(i = 0, 1, \ldots, N-1)$, which represents a network layer (i.e., feature maps). Each edge $e^{(i,j)}$ in DAG represents the information flow from node $x^{(i)}$ to $x^{(j)}$, which consists of the candidate operations. In this article, we denote the operation space as $\mathcal{O}$, which contains 8 candidate operations: 3×3 and 5×5 separable convolutions, 3×3 and 5×5 dilated separable convolutions, 3×3 max pooling, 3×3 average pooling, identity, and zero.

Before searching, the transformations from node $x^{(i)}$ to $x^{(j)}$ are the mixed operations of all candidate operations. The same as DARTS, the operation of each edge is parameterized by architecture parameters $\alpha^{(i,j)}$ as a softmax mixture over all the possible operations. The output of each intermediate node is the sum of information flows from all of its predecessors, and the output of the entire cell is calculated by concatenating the output of nodes $x^0, \ldots, x^{n-1}$.

Owing to the continuous relaxation, the search is realized by optimizing an over-parameterized network. The learning procedure of architecture parameters and weights can be formulated as a bilevel optimization problem shown in (3), (4). Through a first-/second-order approximation [43], the problem can be solved by gradient-based methods. The learning procedure is shown in Fig. 2. Noted that there are two types of cells that require to be learned, normal cell and reduction cell. For the reduction cell, we make the initial operation applied to the cell's inputs have a stride of two to reduce the height and width.

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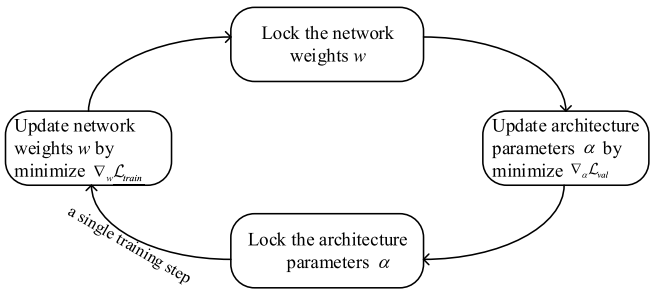


Fig. 2. Gradient-based optimization for architecture parameters and network weights. The idea is to approximate $w^*(\alpha)$ by updating w using only a single training step, without solving the inner optimization (4) completely by training until convergence.

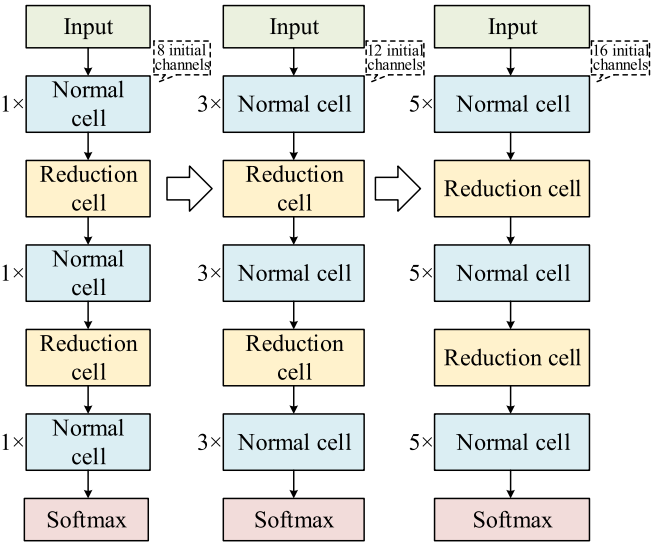


Fig. 3. Whole search process contains three stages. The network depth and width are progressively increased at the end of each stage.

2) GPAS Strategy: As discussed in Section III-A, we consider two issues that arise in DARTS, namely, expensive computational cost and domain-shift problem. In order to address these two issues, the core aspects to reduce are 1) the computational burden and 2) the discrepancy between the search and evaluation phases. In this work, we propose a GPAS framework to find optimal architecture for remote sensing scene classification.

Instead of solving the complete search problem directly, we divide it into subproblems and solve them in a progressive manner. As shown in Fig. 3, the whole search process contains multiple stages and the number of cells increases at the end of each stage, so that after the search phase, the network depth is sufficiently close to the configuration employed in evaluation. In each stage S_k, we denote the number of cells as L_k. For the edges in a cell, the normalized softmax weights of candidate operations can be regarded as a distribution $p_o^{(i,j)} = (\exp(\alpha_o^{(i,j)}))/(\sum_{o' \in \mathcal{O}} \exp(\alpha_{o'}^{(i,j)}))$. We introduce entropy (a common measurement of uncertainty of a distribution) to define a selection criterion for each edge. The selection certainty $SC^{(i,j)}$ of the edge $e^{(i,j)}$ is defined as the complement of the normalized entropy of the operation distribution

$$\mathrm{SC}^{(i,j)} = 1 - \frac{-\sum_{o \in \mathcal{O}} p_o^{(i,j)} \log(p_o^{(i,j)})}{\log(|\mathcal{O}|)}. \tag{5}$$

At the end of each search stage, the architecture parameters α are updated. For the predecessors of each node, the edges with top-k (k is different for different nodes) selection certainty will be selected. We employ an operation-level pruning on the selected edges, where m candidate operations that are less important (with lower scores) are dropped after the search stage. For example, in stage S_k, the intermediate node x_j has e_k^j edges (represent different operations) flow from its previous nodes $x^0, x^1, \ldots, x^{j-1}$. According to the selection certainty, we prune m redundant operations (with lowest scores) in the selected edges after the search process. The architecture parameters α corresponding to the cropped operations are no longer be updated in the next stage. The pruning strategy is illustrated in the lower half of Fig. 1. The path-level pruning reduces the search space progressively and allows us to use a larger batch size for architecture search. In the meantime, we increase the width and depth of the searched architecture by increase the initial channels and stacking more cells. Therefore, the model discrepancy is minimized and the validation accuracy during the search phase reflects the final evaluation accuracy much better.

3) Architecture Regularization and Derivation: Despite the efficiency of progressive search framework, we observe a severe issue in the latter two search stages. That is, after certain search epochs, the number of skip-connections becomes very large because they often leads to rapid convergence, which results in shallow architecture and poor performance. We propose a simple and effective early stopping strategy to avoid the collapse of skip-connections. Specifically, the search procedure should be early stopped when there are two or more than two skip-connects in one normal cell. By this early stopping paradigm, we do not only avoid exploding parameterless operations, which are correlated with poor generalization, but also reduce the search cost. On the other hand, we note that the architecture parameter α of each edge indicates its strength or importance between two nodes, the search procedure aims to rank α within each node based on the score and find the relationship between the candidate operations. When the ranking of α becomes stable, the remaining edges are nearly determined, which indicates that the search is almost saturated. Based on this consideration, we propose a simple architecture regularization scheme. At the initial state, we randomly select mini-batch samples from the training data for learning the architecture parameters with early stopping. and the number of samples and search epochs are increased with the network depth. The purpose of the initial search is to obtain a rough ranking of the architecture parameter. After this search stage, we increase the number of epochs at the next search stage, and more training data are used to learn a refined ranking of α. Such two strategies lead to both better robustness of the ranking of α values and less training cost than the original DARTS.

After the last search stage, we preserve the top-2 strongest nonzero operations (for each intermediate node) aggregated

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from all the previous nodes according to the learned architecture parameters. Then we can stack the searched cells to form a deeper network as the evaluation backbone network. In other words, the architecture of cell is shared by the entire network. The procedure of the proposed methods to search an optimal CNN model for scene classification is shown in Algorithm 1.

Algorithm 1 Framework of the Proposed GPAS Method

1: Initial the architecture parameters $\alpha_{o'}^{(i,j)}(o' \in \mathcal{O})$ for each edges $e^{(i,j)}$ in a cell, the number of cells $L_k(k = 1, 2, 3)$, the number of initial channels $c_k(k = 1, 2, 3)$, the number of pruned edges m, the number of search epoches $n_k(k = 1, 2, 3)$, and the training percentage $r_k(k = 1, 2, 3)$;
2: **for** $k = 1, 2, 3$ (search stage) **do**
3: Initialize the network weights w, learning rate ξ and ε;
4: Set the training percentage to r_k;
5: **for** $i = 0, 1, \ldots, n_k$ **do**
6: Update the architecture parameters α by $\alpha \leftarrow \alpha - \xi \nabla_\alpha \mathcal{L}_{val}(w - \xi \nabla_w \mathcal{L}_{train}(w, \alpha), \alpha)$;
7: Update the network weights w by $w \leftarrow w - \varepsilon \nabla_w \mathcal{L}_{train}(w, \alpha)$;
8: **end for**
9: Prune m redundant operations of the selected edges according the score $p_o^{(i,j)} = \frac{\exp(\alpha_o^{(i,j)})}{\sum_{o' \in \mathcal{O}} \exp(\alpha_{o'}^{(i,j)})}$;
10: Increase the number of cells L_k(network depth) and the number of initial channels c_k(network width);
11: **end for**
12: Output the architecture parameter $\alpha^{(i,j)}$ and derive the final architecture.

IV. Experimental Setup

In this section, the experimental setup for evaluating the proposed methods is presented. At first, Section IV-A gives a detailed description of the used data sets. Then, we construct a new large-scale data set by merging four widely used large-scale remote sensing scene databases. The mergence methods and criteria are described in Section IV-B. In Section IV-C, we present the implementation details of the proposed methods.

A. Data Sets Description

We conduct experimental evaluations on seven public remote sensing scene data sets: AID data set [45], NWPU-RESISC45 (NWPU) data set [29], PatternNet data set [46], RSI-CB data set [47], UC-Merced (UCM) data set [48], RSSCN7 data set [49] and SIRI-WHU data set [50]. These data sets are collected from different satellite sensors under different conditions and over diverse locations of the ground surface, which can cause rich diversities across different data sets. The first four data sets are used as the backbone data set to search the optimal CNN architecture and demonstrate the effectiveness of our methods, and the other three small-scale data sets are employed to evaluate the transferability of searched architecture. A brief introduction to each data set is summarized in Table I and Fig. 4 illustrates some examples of different data sets.

TABLE I
Properties of Different Data Sets

Dataset	Images/class	Classes	Total images	Size	Year
UCM	100	21	2100	256×256	2010
RSSCN7	400	7	2800	400×400	2015
SIRI-WHU	200	12	2400	200×200	2016
NWPU	700	45	31500	256×256	2016
AID	220∼420	30	10000	600×600	2017
RSI-CB	∼690	35	24747	256×256	2017
PatternNet	800	38	30400	256×256	2017



Fig. 4. Sample images of AID (first row), NWPU-RESISC45 (second row), PatternNet (third row), RSI-CB (fourth row), RSSCN7 (fifth row), UCM (sixth row), and SIRI-WHU (last row) remote sensing scene data sets.

B. Construction of a Large-Scale Data Set

Benefiting from the rapid development of remote sensing technologies, more and more large-scale aerial image databases have been released in recent years. The increasing availability of scene data set offers unprecedented opportunities for automatic and accurate interpretation. However, the scale of the existing public data set is still comparatively small and insufficient to search for a suitable and robust CNN architecture for scene classification. In addition, the remote

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sensing images from different data sets are collected by different sensors and over diverse locations of the ground surface, which results in significant discrepancies of semantic information between different databases. The purpose of our work is to automatically search a network architecture with satisfactory generalization performance and transferability. Therefore, we propose a mergence strategy to build a new large-scale scene image data set, referred to as VHRRS data set. In our method, four heterogeneous and comparatively large-scale remote sensing image data sets, AID, NWPU, PatternNet, and RSI-CB are selected to construct the VHRRS data set. Based on the idea of multiduplication and multicomplementary, we merge the sample images from different data sets according to the following rules.

1) Since different data sets are annotated by different experts, the same scene from different data sets may be labeled with different class names. For example, the class "viaduct" in AID is labeled with "overpass" in NWPU and PatternNet, and the class "crossroad" in RSI-CB is labeled with "intersection" in NWPU and PatternNet. In our work, new uniform class names are used to merge the same scenes from different data sets.
2) Due to different annotation criteria, some different scene categories from different data sets may contain the same surface objects. For instance, some images in the class "artificial grassland" in RSI-CB may contain "football filed" or "baseball filed", while these two scenes are labeled, respectively, with "football field" in PatternNet and "baseball diamond" in NWPU. To address this issue, we reclassify and remerge the sample images of different categories that contain overlapping scene information.
3) Due to some ambiguous category definitions in different data sets, we remove some sample images that contain uncertain geometrical structures of scenes. For example, the class "city building" in RSI-CB can also be labeled with "commercial area" in NWPU.

Based on the above rules, we construct a new large-scale remote sensing image data set, which consists of 93 068 remote sensing images with 81 scene classes. Noted that the merged data set contains multiple source-specific domain information of scene representation, which is helpful to search robust network architecture with powerful generalized capability.

C. Implementation Details

To evaluate the effectiveness of the proposed methods for scene classification, we first employ the search algorithm to find an optimal CNN architecture on the merged data set VHRRS. After the optimal architecture is obtained, the searched architecture is, respectively, transferred to four large-scale benchmark data sets (i.e., AID, NWPU, PatternNet, and RSI-CB) through an individual training process from scratch. In addition, we evaluate the performance of searched architecture on the other three small-scale data sets to demonstrate its transferability and generalization capability. Finally, the necessity and effectiveness of the cross-domain merged large-scale data set are investigated.

For the merged data set VHRRS, we split the sample images into three subsets based on stratified sampling, i.e., 40% samples for training, 20% samples for validation, and 40% samples for test. In the search phase, the training and validation data sets are used to learn the parameter α. After the optimal architecture is determined, the searched architecture is transferred to four individual benchmark data sets (i.e., AID, NWPU, PatternNet, and RSI-CB) through full-training using the samples used for search. Note that the training samples used in the search phase are never used in the evaluation of the transferred architecture.

According to the proposed greedy and progressive search strategy, the whole search phase is divided into three stages. The number of cells in the three stages is set to 5, 11, and 17, respectively. The number of initial channels (i.e., the network width) in each stage is set to 8, 12, and 16, respectively. Each cell contains $N = 7$ nodes (i.e., two input nodes, four intermediate nodes and one output node). Based on the edge selection criterion, the number of selected cropped edges for the intermediate node is set to 1, 2, 2, and 3, respectively. According to the architecture regularization methods, the search epoch in each stage is set to 15, 25, and 30, while the utilization percentage of training data (include sampled training and validation data) for different stages is set to 60%, 80%, and 100%, so that we can progressively obtain more refined ranking of parameter α.

In the search phase, all of the training images are resized to a fixed size of 64×64 pixels. We perform common data augmentation technologies on training, including random cropping, rotation and flip to avoid over-fitting and ensure the robustness of the searched architecture. The experiments are implemented with Pytorch [51] framework using NVIDIA GTX 1080Ti GPUs. Considering the memory cost, we train the network using a batch size of 32 for each search stage. An Adam optimizer with learning rate $\eta = 0.0005$, weight decay $= 0.001$ and momentum $\beta = (0.5,\ 0.999)$ is employed for learning architecture parameters α. We use SGD to optimize the model weights with initial learning rate 0.025, momentum 0.9 and weight decay 3×10^{-4}.

In the evaluation phase, the number of preserved operations for each node is set to 2. The learned optimal cell is stacked to construct the final architecture. We build a network of 20 cells and 32 initial channels for the evaluation on four large-scale data sets (i.e., AID, NWPU, PatternNet, and RSI-CB), respectively. We train the final model from scratch for 500 epochs with batch size 64 to ensure convergence. Other hyperparameters and configurations remain the same as the ones used in the search phase. Following previous works [52], [53], we also employ two simple enhancement strategies including cutout regularization [54] of length 16 and path dropout with probability 0.2.

For the experimental results, we employ two metrics for quantitative evaluation, including the overall accuracy (OA) and the mean Average Precision (mAP).

V. Results and Analysis

In this section, we report the experimental results of the proposed methods for remote sensing scene classification tasks.

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TABLE II
GPU MEMORY USAGE AND SEARCH TIME AT DIFFERENT STAGES DURING THREE INDIVIDUAL RUNS

Run No.	Memory usage(GB)			Search time(h)		
	Stage 1	Stage 2	Stage 3	Stage 1	Stage 2	Stage 3
1	9.5	14.2	14.3	4.0	8.2	9.5
2	9.5	14.5	14.1	4.2	8.0	9.2
3	9.5	14.3	14.2	4.0	8.1	9.2

At first, we give an analysis for the search space, and then the architecture search visual results are presented. Next, the performance of searched cells is evaluated on four large-scale data sets and three small-scale data sets, respectively, and the comparison results with other methods are also reported. Finally, we analyze the transferability of the searched architecture and the effectiveness of the merged cross-domain data set.

A. Analysis of Search Space

In this section, we analyze the complexity of the search space for our architecture search framework. The initial search space in our experiments is the same as that used in DARTS, which contains 8 candidate operations, i.e., $\mathcal{O}$ ={3×3 *separable convolution*, 3×3 *dilated separable convolution*, 5×5 *separable convolution*, 5×5 *dilated separable convolution*, 3×3 *max pooling*, 3×3 *average pooling*, *identity*, *zero*}. Each cell contains $N = 7$ nodes, where the first two nodes are equal to the outputs of the previous two cells, respectively, and the output node is computed by the concatenation of all the intermediate nodes. Theoretically, each discretized cell allows $\prod_{k=1}^{4}(k(k+1)/2) \times (7^2) \approx 10^9$ possible DAGs without considering graph isomorphism. Since two types of cells are jointly learned during the search phase, the total number of architectures is approximately $(10^9)^2 = 10^{18}$. Note that we preserve the top-2 strongest nonzero operations (for each intermediate node) after the search phase, and the continuous search space before this final derivation step is even larger. Specifically, each relaxed cell contains $2 + 3 + 4 + 5 = 14$ learnable edges, allowing $(7 + 1)^{14} \approx 4 \times 10^{12}$ possible configurations ($+1$ to include the zero operation indicating a lack of connection). Again, since there are two types of cells are learned, the total number of possible architectures covered by the search space before derivation is $(4 \times 10^{12})^2 \approx 10^{25}$. Our search strategy allows candidate operations for each edge to decrease as the depth increases, which means the number of learnable architecture parameters and the search space become smaller at the end of each search stage.

The search process on VHRRS data set takes approximately 21.5 h (1.8 days) on two Nvidia GTX 1080Ti GPUs to produce the final architectures. We report GPU memory usage and search cost of each search stage for three individual runs, which are shown in Table II. At stage 1, we can see that the memory usage and search time are less than that in next two stages due to the shallow depth. Although the learnable architecture parameters are decreased, the number of model weights is significantly increased with the network depth. The memory usage in different runs is stable, which shows the search process is stable and proves the effectiveness of the progressive search strategy for memory efficiency.

B. Search Results

In our experiments, we, respectively, employ GPAS and DARTS to search the optimal network architecture on the VHRRS data set, and then we compare the cell architectures searched by our methods with that discovered by DARTS. The detailed search results are illustrated in Fig. 5. Two types of cells are searched during the search phase, namely normal cell and reduction cell. The reduction cell is inserted at a network depth of 1/3 and 2/3, respectively. The stride of convolution operation in normal cells is 1, so the size of the output feature map does not change. In reduction cells, the operation stride in the edges connected with the input nodes is set to 2 in order to reduce the spatial resolution of feature maps. Compared with DARTS, we can find that the cells searched by GPAS tend to preserve more deep connections (e.g., convolutions) rather than shallow connections (e.g., skip connection), which contains less learnable parameters than deep ones, thus it has a weaker expressive ability. In other words, our method learns more layers in the cell, making the final network further deeper and performs better.

C. Evaluation Results

Through stacking the searched cell structures, we obtain the final network for evaluations. In the experiments, we investigate the representative ability of the final network and evaluate the classification performance on four large-scale benchmark data sets. We conduct two sets of comparison experiments, where the first set of comparison methods are performed on classical CNN models (with and without pretraining) to verify the limitations of the pretraining model and the effectiveness of automatic designed CNN architecture for scene classification. Three representative CNN models are selected for comparison, including VGG-16 [22], GoogLeNet [23], and ResNet-50 [55], all of which are usually employed as the feature extractor in various CNN-based methods. We set the proportion of training samples in the AID data set, NWPU data set, RSI-CB data set, and PatternNet data set to 50%, 60%, 60%, and 50%, respectively. The classification results on four large-scale data sets are shown in Table III, where the second to fourth rows list the classification results of three classical CNN models without any pretraining. In this case, the network weights are randomly initialized and trained from scratch. Regardless of OA or mAP, we can find that the classification accuracies on AID and NWPU data sets are below 90%, which are poor and unsatisfactory. However, the scale of other two data sets PatternNet and RSI-CB are relatively large and the patterns are quite easy to recognize for these CNN architectures with a majority of the classifications reaching 96% OA, which also reflects the importance of the amount of training data. The next three rows list the results based on pretrained models. For a fair comparison, the models are initialized with pretrained weights based on ImageNet and then fine-tuned on the target data

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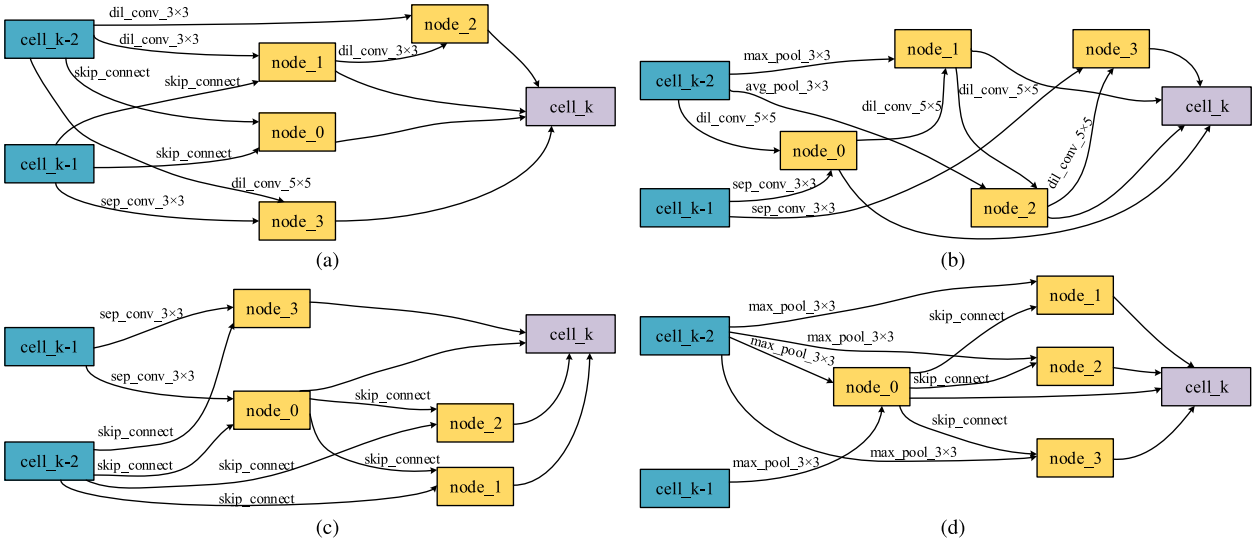


Fig. 5. Searched cell architectures of two methods on VHRRS data set. (a) Normal cell searched by GPAS. (b) Reduction cell searched by GPAS. (c) Normal cell searched by DARTS. (d) Reduction cell searched by DARTS.

TABLE III

CLASSIFICATION RESULTS OF THE PROPOSED METHODS AND OTHER CNN-BASED METHODS ON FOUR LARGE-SCALE DATA SETS

	AID		NWPU		RSI-CB		PatternNet	
	mAP(%)	OA(%)	mAP(%)	OA(%)	mAP(%)	OA(%)	mAP(%)	OA(%)
full-trained VGG-16	87.98	88.56	85.29	85.97	95.23	96.53	96.23	97.31
full-trained GoogLeNet	84.23	84.27	83.03	83.29	96.02	96.21	95.81	96.12
full-trained ResNet-50	86.21	86.94	85.98	86.96	97.05	97.09	95.97	96.71
VGG-16 (pretraining+fine-tuning)	91.58	92.03	89.93	91.32	97.02	97.26	98.23	98.31
GoogLeNet (pretraining+fine-tuning)	89.69	90.31	88.20	89.42	98.10	98.14	97.52	97.56
ResNet-50 (pretraining+fine-tuning)	91.98	92.14	90.92	91.63	97.52	98.12	97.95	98.23
the searched models (40% training samples)	94.52	93.85	92.95	92.57	97.99	99.01	98.35	98.25
the searched models (60% training samples)	**95.08**	**95.28**	**93.60**	**94.87**	**98.69**	**99.04**	**98.95**	**99.13**

set. As expected, the pretrained CNNs consistently achieve significant improvement compared with the full-trained models, which indicates the powerful expressive and discriminative ability of pretrained models. The classification results obtained by our methods are shown in the last row. For the data set AID and NWPU, the searched CNN architecture achieves the best performance on both OA and mAP, with a significant accuracy improvement of 7∼11% over the classical full-trained CNN models. Simultaneously, the classification performance of our model is also better than the manually designed CNN architectures with fine-tuning and it results in 3%–5% accuracy improvement compared to three pretrained models. For the data set PatternNet and RSI-CB, it also gains similar classification accuracy with that obtained by pretrained model-based methods. In addition, we decrease the number of training samples to better analyze the effects of the training percentages. As expected, in the case of fewer training samples (40% training samples), the classification accuracy of our method on the AID data set is higher than full-trained VGG-16, GoogLeNet, and ResNet-50 by 5.29%, 9.58%, and 6.91%, respectively. We can conclude that the pretrained CNN-based methods do not always provide better generalization, and are not the optimal choice for scene classification due to the limited training data as well as the significant discrepancies between the natural and remote sensing images. In addition, the NAS-based methods are effective and promising, which can learn specific semantic features on the target task and automatically search suitable network architecture.

In the second set of comparison experiments, the efficiency and robustness of the proposed GPAS are verified in the task of architecture search for scene classification. Four state-of-the-art NAS methods are used for comparison, including ENAS [53], DARTS [43], ProxylessNAS [52], and P-DARTS [44]. Standard NAS paradigm is conducted in our experiments, i.e., searching the optimal architecture on the merged large-scale data set VHRRS, and evaluating on two benchmark data sets AID and NWPU, respectively. Moreover, we report the classification results of a representative manually designed CNN model DenseNet-169 [56], which is one of the best-performing architectures for image classification. For the selected model, we follow the experimental configurations in the previous works [52], [53], [43] for evaluation on different data sets. Table IV summarizes the evaluation results of two data sets, including OA, search cost (the number of GPU days)

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TABLE IV
COMPARISON RESULTS WITH STATE-OF-THE-ART NAS-BASED METHODS ON AID AND NWPU DATA SETS

	Overall Accuracy(%)		Parameters (M)		Search cost	Search strategy
	AID	NWPU	AID	NWPU	(GPU-days)	
DenseNet-169	91.86	90.36	28.5	28.8	-	Manually designed
ENAS	93.89	93.54	4.8	4.7	3.0	Reinforcement Learning
DARTS	94.24	93.04	**3.4**	**3.4**	4.5	Gradient-based
ProxylessNAS	94.57	93.23	6.2	6.2	6.5	Gradient-based
P-DARTS	94.84	94.09	3.5	3.5	2.0	Gradient-based
GPAS	95.28	94.87	3.6	3.7	1.8	Gradient-based
GPAS + cutout	**96.35**	**95.16**	3.6	3.7	**1.8**	Gradient-based

and the number of parameters. For the classification performance on either AID or NWPU, we can find that the proposed GPAS achieves the highest accuracy, reaching 95.28% and 94.87%, respectively, which are much better than full-training DenseNet that is designed by human experts. In addition, GPAS also outperforms the other four NAS-based methods on classification performance, including a representative RL-based method and three efficient gradient-based methods. In terms of the computation resources, our method only costs 1.8 GPU days to find a competitive architecture and the number of parameters (3.6 M) is far less than that in DenseNet (28.5 M). To obtain a similar performance, ProxylessNAS spent the longest search time which is approximately four times slower than GPAS. Our search methods also outperform two other approaches DARTS and P-DARTS, which are known for search efficiency. Moreover, the classification accuracy achieves further improvements by incorporating cutout regularization strategy, reaching 96.35% and 95.16%, respectively. The remarkable empirical results of our GPAS demonstrate the benefits of progressively increase the network depth and refine the search space. The proposed strategies reduce the discrepancy between the search and evaluation phase, and make the search process more efficient.

D. Transferability of Searched Architecture

To evaluate the transferability of CNN architecture discovered by our methods, we investigate the performance of the searched cell architecture on the three small-scale data sets, i.e., UCM, RSSCN7, and SIRI-WHU which are not involved in the search process. The images from the three benchmark data sets are acquired with various sensors under different conditions (e.g., the training images and target images vary greatly in shooting angles, spatial, and spectral resolutions), which cause large distribution discrepancies across different data sets. To avoid over-fitting, the evaluation network for the small-scale data sets only consists of 12 cells with 32 initial channels and is trained from scratch for a total of 500 epochs. The same as the experiments above, three classical CNN models with randomly initialized weights (i.e., VGG-16, GoogLeNet, and ResNet-50) are fully trained for comparison. Follow the experimental setup in [11], [45], the proportion of training samples in UCM data set, RSSCN7 data set, and SIRI-WHU data set is set to 80%, 50%, and 50%, respectively.

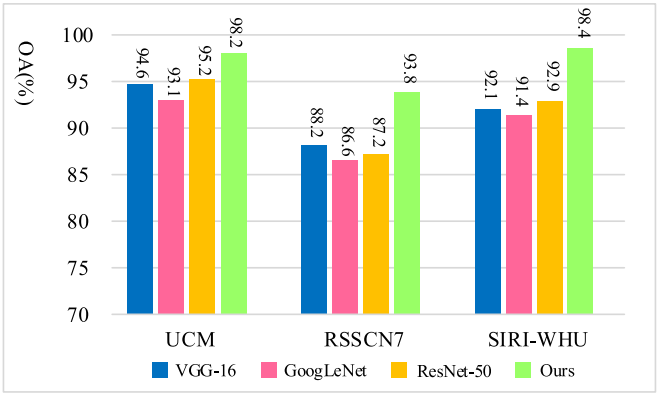


Fig. 6. Classification results of three small-scale data sets.

Fig. 6 shows the classification results with different full-training CNN models. As excepted, the proposed GPAS is more suitable to discover a network architecture that provides discriminative feature representations for aerial scene classification. The proposed method achieves the best classification results with an OA of 98.2%, 93.8%, and 98.4% on UCM, RSSCN7, and SIRI-WHU data set, respectively, which are higher than all of the comparison methods based on classical CNN models. We infer that the significant differences between natural and aerial images cause the reduced discrimination of conventional manually designed CNN architectures. Although the semantic features of the target domain are not learned in the search process, the architecture search method can capture the cross-domain similarities and generalized feature representations of aerial images. It can be concluded that the architectures searched by the proposed GPAS achieve impressive transferability and reduce the domain-shift on the task of scene classification.

E. Effectiveness of Cross-Domain Large-Scale Data Set

In this section, we demonstrate the necessity and effectiveness of the cross-domain large-scale data set VHRRS that is a combination of four subdata sets based on a specific mergence strategy. We perform two sets of comparative experiments on three data sets AID, NWPU, and UCM. The first set of experiments is similar to the experiment above, where the optimal cells searched on the merged data set VHRRS. The searched cells are stacked to form a complete network,

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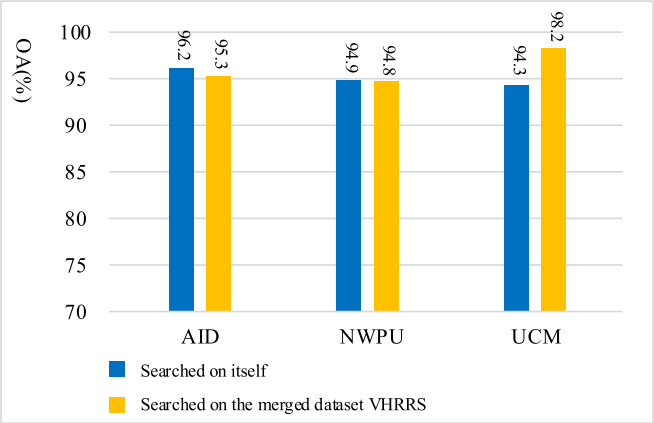


Fig. 7. Comparison results with different search schemes.

which is then full-trained on the three data sets, respectively. In the second set of experiments, the proposed architecture search method GPAS is employed to find suitable architecture on three small data sets, respectively. In other words, each data set itself, rather than the merged data set VHRRS, is used as the backbone data set, respectively. The classification results are shown in Fig. 7. For the two relatively large-scale data sets AID and NWPU, it can be seen that the architecture searched on themselves achieves competitive classification accuracies, reaching 96.2% and 94.9%, respectively, which are slightly higher than that discovered on VHRRS data set. However, the classification results of the UCM data set show a relatively large gap between the two search strategies. The architecture learned by VHRRS data set achieves satisfactory performance, reaching 98.2%, which is much better than that searched on UCM. We infer that the UCM data set could be lack of image variations and diversity and the training data is limited. Through the results, we conclude that the number of labeled training data has a significant effect on the search performance, sufficient training data and rich image diversity are crucial to find a suitable architecture. Therefore, the union of knowledge from multiple-source domains is necessary to generate a new CNN architecture with powerful generalization capability.

VI. Conclusion

In this article, we investigate a new paradigm to automatically design a suitable CNN architecture for scene classification due to the limitations of the features extracted from the pretrained CNN models and the deficiency of available scene image data. We propose an efficient architecture search framework, where the network architectures are discovered through a greedy and progressive search strategy with gradient-based optimization. It effectively alleviates the domain-shift between search and evaluation phases. In addition, we propose a simple architecture regularization scheme to further improve the search efficiency and the robustness of discovered architectures. Moreover, we conduct a new large-scale scene data set, which combines multiple public scene data sets with different domains. Extensive experiments demonstrate the efficiency of the proposed search strategies, and the impressive classification performance of searched CNN architectures on seven public benchmark data sets.

Our works open up a new path toward various tasks of remote sensing image analysis. We believe that the potential of NAS methods in the field of remote sensing will receive increasing interests. The future works include research on more efficient NAS methods to adapt more complex remote sensing image interpretation tasks.

References

[1] X. X. Zhu *et al.*, "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.
[2] K. Nogueira, O. A. B. Penatti, and J. A. dos Santos, "Towards better exploiting convolutional neural networks for remote sensing scene classification," *Pattern Recognit.*, vol. 61, pp. 539–556, Jan. 2017.
[3] F. Hu, G.-S. Xia, J. Hu, and L. Zhang, "Transferring deep convolutional neural networks for the scene classification of high-resolution remote sensing imagery," *Remote Sens.*, vol. 7, no. 11, pp. 14680–14707, Nov. 2015.
[4] X. Lu, Y. Yuan, and X. Zheng, "Joint dictionary learning for multispectral change detection," *IEEE Trans. Cybern.*, vol. 47, no. 4, pp. 884–897, Apr. 2017.
[5] Y. Li, C. Peng, Y. Chen, L. Jiao, L. Zhou, and R. Shang, "A deep learning method for change detection in synthetic aperture radar images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5751–5763, Aug. 2019.
[6] O. E. Dai, B. Demir, B. Sankur, and L. Bruzzone, "A novel system for content-based retrieval of single and multi-label high-dimensional remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 7, pp. 2473–2490, Jul. 2018.
[7] C. Peng, Y. Li, L. Jiao, Y. Chen, and R. Shang, "Densely based multi-scale and multi-modal fully convolutional networks for high-resolution remote-sensing image semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 2612–2626, Aug. 2019.
[8] P. Ghamisi *et al.*, "New frontiers in spectral-spatial hyperspectral image classification: The latest advances based on mathematical morphology, Markov random fields, segmentation, sparse representation, and deep learning," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 3, pp. 10–43, Sep. 2018.
[9] Z. Wang *et al.*, "Visual attention-based target detection and discrimination for high-resolution SAR images in complex scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1855–1872, Apr. 2018.
[10] X. Zheng, Y. Yuan, and X. Lu, "A deep scene representation for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4799–4809, Jul. 2019.
[11] Q. Zhu, Y. Zhong, L. Zhang, and D. Li, "Adaptive deep sparse semantic modeling framework for high spatial resolution image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6180–6195, Oct. 2018.
[12] F. Hu, G.-S. Xia, W. Yang, and L. Zhang, "Recent advances and opportunities in scene classification of aerial images with deep models," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Jul. 2018, pp. 4371–4374.
[13] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.
[14] L.-J. Zhao, P. Tang, and L.-Z. Huo, "Land-use scene classification using a concentric circle-structured multiscale bag-of-visual-words model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 12, pp. 4620–4631, Dec. 2014.
[15] X. Lu, X. Zheng, and Y. Yuan, "Remote sensing scene classification by unsupervised representation learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5148–5157, Sep. 2017.
[16] J. Sánchez, F. Perronnin, T. Mensink, and J. Verbeek, "Image classification with the Fisher vector: Theory and practice," *Int. J. Comput. Vis.*, vol. 105, no. 3, pp. 222–245, Dec. 2013.
[17] M. Castelluccio, G. Poggi, C. Sansone, and L. Verdoliva, "Land use classification in remote sensing images by convolutional neural networks," 2015, *arXiv:1508.00092*. [Online]. Available: http://arxiv.org/abs/1508.00092
[18] F. P. S. Luus, B. P. Salmon, F. van den Bergh, and B. T. J. Maharaj, "Multiview deep learning for land-use classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 12, pp. 2448–2452, Dec. 2015.
[19] B. Li *et al.*, "Aggregated deep Fisher feature for VHR remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3508–3523, Sep. 2019.

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[20] Y. Bazi, M. M. Al Rahhal, H. Alhichri, and N. Alajlan, "Simple yet effective fine-tuning of deep CNNs using an auxiliary classification loss for remote sensing scene classification," *Remote Sens.*, vol. 11, no. 24, p. 2908, Dec. 2019.
[21] Y. Jia *et al.*, "Caffe: Convolutional architecture for fast feature embedding," in *Proc. ACM Int. Conf. Multimedia - MM*, 2014, pp. 675–678.
[22] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*. [Online]. Available: http://arxiv.org/abs/1409.1556
[23] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.
[24] E. Li, J. Xia, P. Du, C. Lin, and A. Samat, "Integrating multilayer features of convolutional neural networks for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5653–5665, Oct. 2017.
[25] K. He, R. Girshick, and P. Dollar, "Rethinking ImageNet pre-training," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4918–4927.
[26] B. Zoph, V. Vasudevan, J. Shlens, and Q. V. Le, "Learning transferable architectures for scalable image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8697–8710.
[27] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
[28] N. He, L. Fang, S. Li, A. Plaza, and J. Plaza, "Remote sensing scene classification using multilayer stacked covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 6899–6910, Dec. 2018.
[29] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.
[30] W. Zhao and S. Du, "Scene classification using multi-scale deeply described visual words," *Int. J. Remote Sens.*, vol. 37, no. 17, pp. 4119–4131, Sep. 2016.
[31] F. Zhang, B. Du, and L. Zhang, "Scene classification via a gradient boosting random convolutional network framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1793–1802, Mar. 2016.
[32] Y. Liu and C. Huang, "Scene classification via triplet networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 220–237, Jan. 2018.
[33] Q. Zhu, Y. Zhong, Y. Liu, L. Zhang, and D. Li, "A dep-local-gobal feature fusion framework for high spatial resolution imagery scene classification," *Remote Sens.*, vol. 10, no. 4, p. 568, Apr. 2018.
[34] B. Zhao, Y. Zhong, and L. Zhang, "A spectral–structural bag-of-features scene classifier for very high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 116, pp. 73–85, Jun. 2016.
[35] P. Du, E. Li, J. Xia, A. Samat, and X. Bai, "Feature and model level fusion of pretrained CNN for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 2600–2611, Aug. 2019.
[36] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.
[37] R. Zhu, L. Yan, N. Mo, and Y. Liu, "Attention-based deep feature fusion for the scene classification of highresolution remote sensing images," *Remote Sens.*, vol. 11, no. 17, p. 1996, Aug. 2019.
[38] W. Han, R. Feng, L. Wang, and Y. Cheng, "A semi-supervised generative framework with deep learning features for high-resolution remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 23–43, Nov. 2018.
[39] I. Bello, B. Zoph, V. Vasudevan, and Q. V. Le, "Neural optimizer search with reinforcement learning," in *Proc. 34th Int. Conf. Mach. Learn.*, vol. 70, 2017, pp. 459–468.
[40] B. Baker, O. Gupta, N. Naik, and R. Raskar, "Designing neural network architectures using reinforcement learning," 2016, *arXiv:1611.02167*. [Online]. Available: http://arxiv.org/abs/1611.02167
[41] B. Zoph and Q. V. Le, "Neural architecture search with reinforcement learning," 2016, *arXiv:1611.01578*. [Online]. Available: http://arxiv.org/abs/1611.01578
[42] E. Real, A. Aggarwal, Y. Huang, and Q. V. Le, "Regularized evolution for image classifier architecture search," in *Proc. aaai Conf. Artif. Intell.*, vol. 33, 2019, pp. 4780–4789.
[43] H. Liu, K. Simonyan, and Y. Yang, "DARTS: Differentiable architecture search," 2018, *arXiv:1806.09055*. [Online]. Available: http://arxiv.org/abs/1806.09055
[44] X. Chen, L. Xie, J. Wu, and Q. Tian, "Progressive differentiable architecture search: Bridging the depth gap between search and evaluation," 2019, *arXiv:1904.12760*. [Online]. Available: http://arxiv.org/abs/1904.12760
[45] G.-S. Xia *et al.*, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[46] W. Zhou, S. Newsam, C. Li, and Z. Shao, "PatternNet: A benchmark dataset for performance evaluation of remote sensing image retrieval," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 197–209, Nov. 2018.
[47] H. Li *et al.*, "RSI-CB: A large scale remote sensing image classification benchmark via crowdsource data," 2017, *arXiv:1705.10450*. [Online]. Available: http://arxiv.org/abs/1705.10450
[48] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst. GIS*, 2010, pp. 270–279.
[49] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.
[50] B. Zhao, Y. Zhong, G.-S. Xia, and L. Zhang, "Dirichlet-derived multiple topic scene classification model for high spatial resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2108–2123, Apr. 2016.
[51] A. Paszke *et al.*, "Automatic differentiation in PyTorch," in *Proc. 31st Conf. Neural Inf. Process. Syst. (NIPS) Workshop*, Long Beach, CA, USA, Dec. 2017.
[52] H. Cai, L. Zhu, and S. Han, "ProxylessNAS: Direct neural architecture search on target task and hardware," 2018, *arXiv:1812.00332*. [Online]. Available: http://arxiv.org/abs/1812.00332
[53] H. Pham, M. Y. Guan, B. Zoph, Q. V. Le, and J. Dean, "Efficient neural architecture search via parameter sharing," 2018, *arXiv:1802.03268*. [Online]. Available: http://arxiv.org/abs/1802.03268
[54] T. DeVries and G. W. Taylor, "Improved regularization of convolutional neural networks with cutout," 2017, *arXiv:1708.04552*. [Online]. Available: http://arxiv.org/abs/1708.04552
[55] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[56] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.



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